



Master in Computer Science
Software Engineering

Cloud Computing

Report 4 **Team Members**

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Exercise 1. S3 Bucket Creation and Upload Objects

Question 1. Create an S3 Bucket for File Storage

We created an S3 bucket in the AWS Management Console with the following configuration:

Bucket Settings:

- **Name:** bdiacc-group3
- **Region:** Europe (Paris) eu-west-3
- **Object Ownership:** ACLs disabled
- **Block Public Access:** Enabled
- **Versioning:** Disabled
- **Encryption:** SSE-S3 enabled

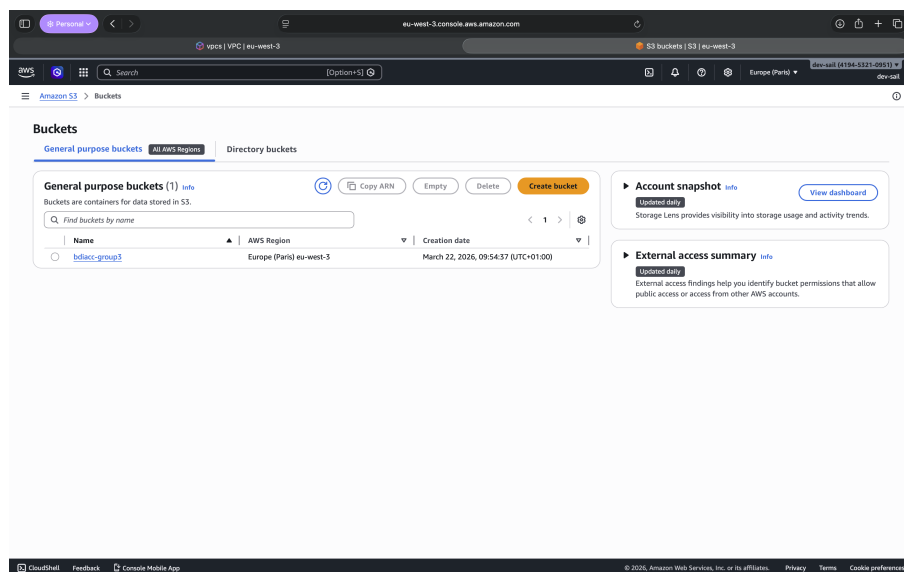


Figure 1: S3 bucket created

Question 2. Upload 5 Objects with Different File Extensions

We uploaded 5 different file types to the bucket to demonstrate how S3 handles various formats:

- `requirement.txt` - 384.0 B
- `ch00.pdf` - 142.7 KB
- `main.py` - 2.4 KB
- `setup.md` - 5.6 KB
- `favicon.ico` - 15.0 KB

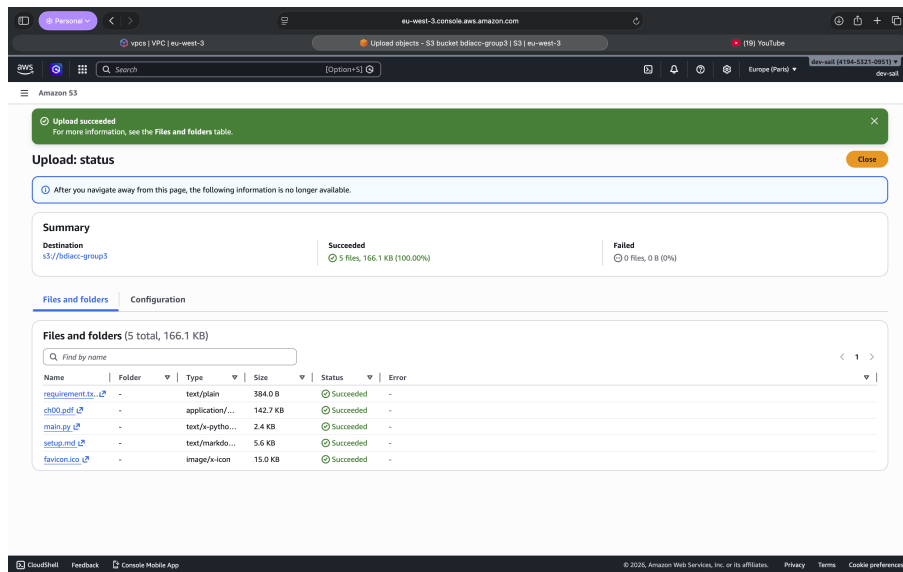


Figure 2: Objects uploaded to S3 bucket

All files uploaded correctly using the drag-and-drop interface.

Exercise 2. S3 Static Website Hosting

Question 1. Enable Static Website Hosting

We enabled static website hosting on the bucket. This allows the bucket to serve HTML content directly over HTTP.

Configuration:

- **Index document:** index.html
- **Error document:** error.html

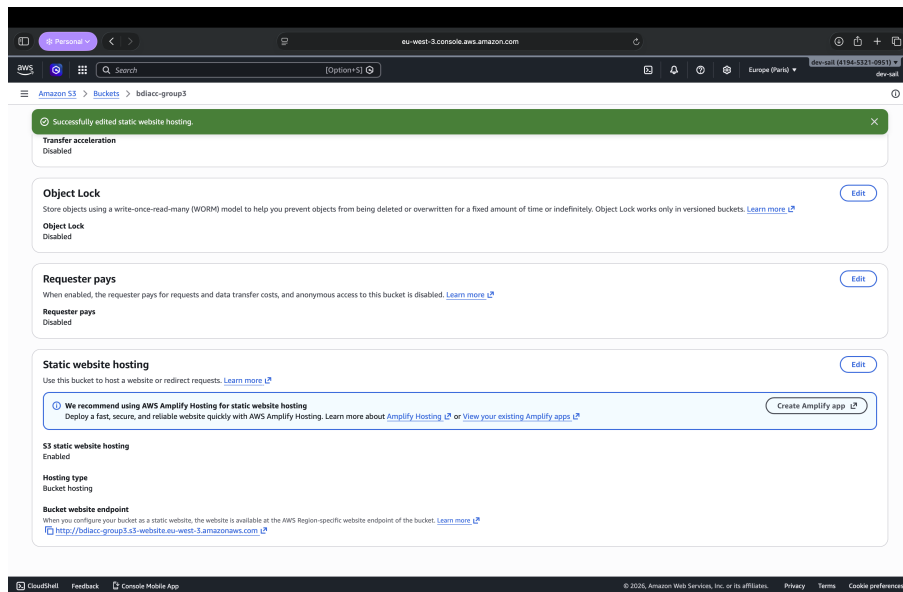


Figure 3: Static website hosting enabled

The endpoint provided by AWS is:

`http://bdiacc-group3.s3-website.eu-west-3.amazonaws.com`

Question 2. Upload index.html and error.html Files

We uploaded HTML files to support the website. The index.html page shows group information:

- S3 Bucket: bdiacc-group3
- Group 3
- Big Data Infrastructure and Cloud Computing
- MSc Computer Science, Software Engineering
- Team: Jordy Meye, Huizhe Su, Arihant Jain, Nizar

The error.html displays a 404 page with a "Go back" link for users who request missing pages.

Question 3. Configure Bucket Policy for Public Access

When we first tried to access the website, we got a 403 Forbidden error because the bucket was set to block all public access.

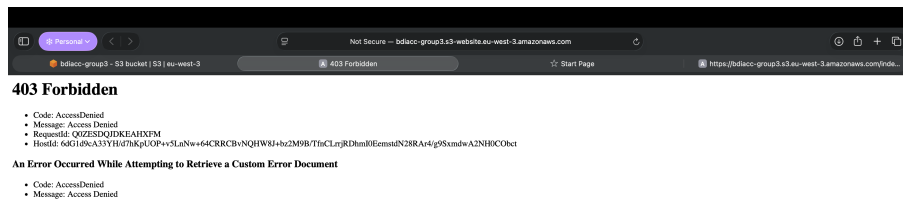


Figure 4: 403 error before bucket policy

We added a bucket policy to allow public read access:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::bdiacc-group3/*"
    }
  ]
}
```

Question 4. View HTML Files

After applying the policy, both files became accessible through the S3 website endpoint.

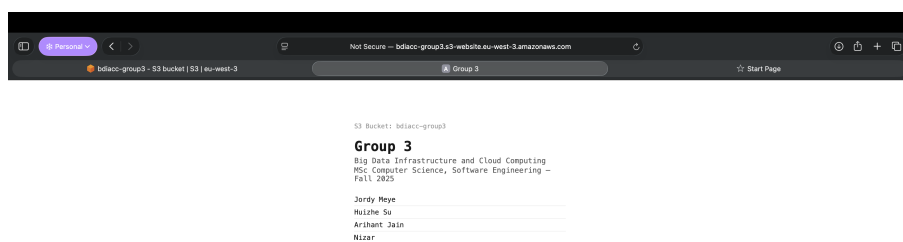


Figure 5: index.html

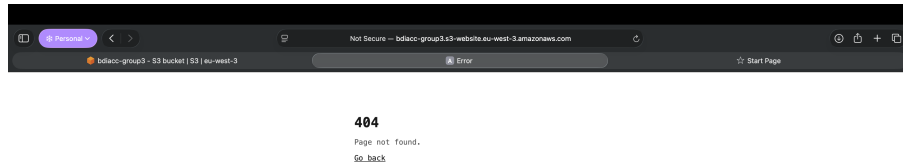


Figure 6: error.html

Question 5. Add Lifecycle Rules

We configured a lifecycle rule named "StorageClassChange" with the following settings:

- Day 60: Move objects to Standard-IA storage class
- Day 200: Delete objects
- For noncurrent versions: Keep 2 newest versions, move others to Standard-IA at day 30

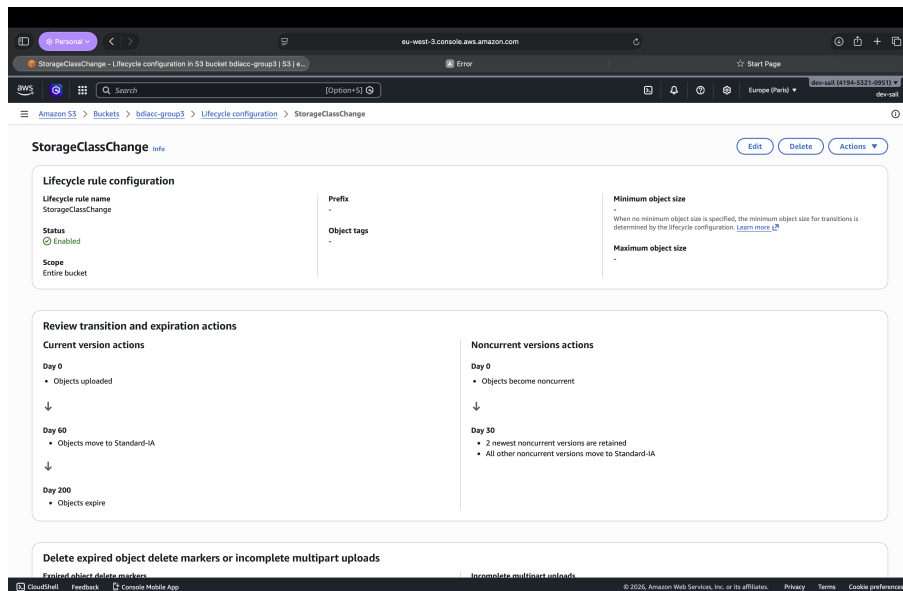


Figure 7: Lifecycle rule configuration

This reduces storage costs by automatically moving old objects to cheaper storage or deleting them.

Exercise 3. S3 Bucket Versioning

Question 1. Enable Versioning for the Bucket

We enabled versioning on the bucket so that previous versions of files are kept when they are updated.

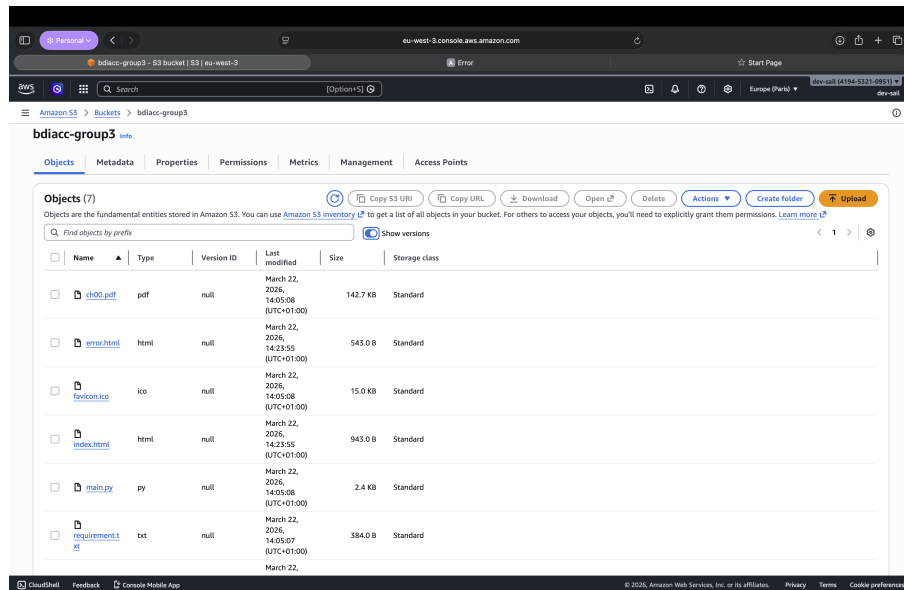


Figure 8: Versioning enabled

Once enabled, every object gets a unique version ID. Old versions can be restored if needed.

Question 2. Re-upload Files and Verify Versioning

We uploaded new versions of index.html and error.html to test how versioning works. S3 keeps both the old and new versions.

Original versions:

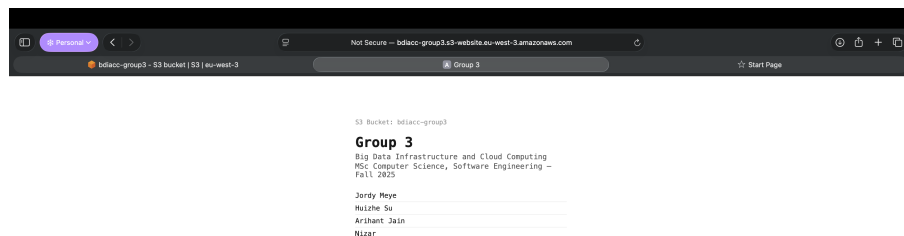


Figure 9: Original index.html

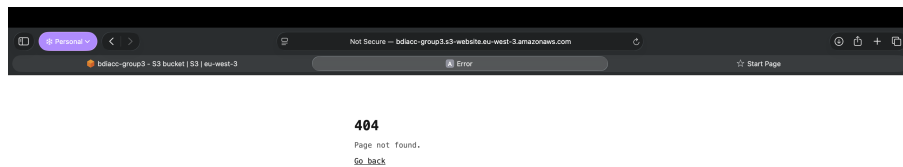


Figure 10: Original error.html

Updated versions:

When we re-uploaded the files, they got new version IDs:

- index.html: Xo5opup2H_I9L5eHRetu_WswrJkVmNto
- error.html: .0ZCYhTJt8YmBbY2G3ANgHptOzaMKrDV

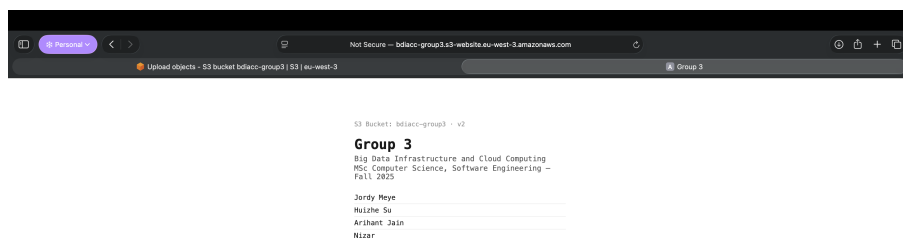


Figure 11: Updated index.html

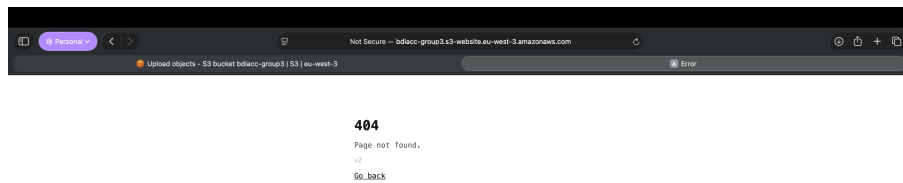


Figure 12: Updated error.html

Question 3. Verify Multiple Versions in Bucket

The bucket now contains multiple versions of each file. In the S3 console, both old and new versions are visible.

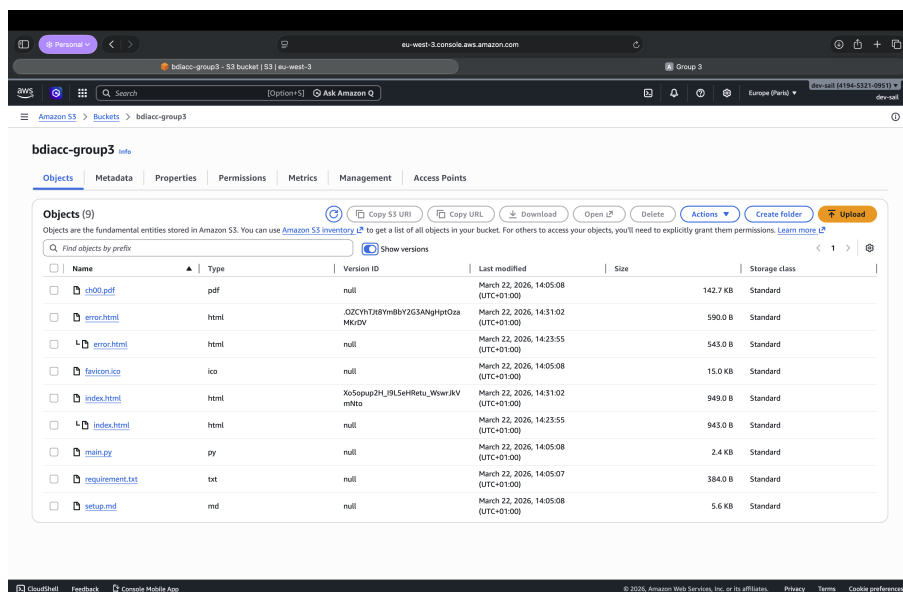


Figure 13: Bucket with multiple file versions

The bucket contains:

- 9 objects total
- index.html: Version ID Xo5opup2H_I9L5eHRetu_WswrJkVmNto (949.0 B, 14:31:02) and older version (943.0 B, 14:23:55)
- error.html: Version ID .OZCYhTJt8YmBbY2G3ANgHpt0zaMKrDV (590.0 B, 14:31:02) and older version (543.0 B, 14:23:55)
- Other files: ch00.pdf, favicon.ico, main.py, requirement.txt, setup.md

Versioning allows us to restore older versions if a new upload causes problems.